

Pure Mathematics (A-Level)

G1: Algebra

Algebra – P1

1. Background Algebra

- Manipulating Algebraic Expression
 - You will often wish to tidy up an expression, or to rearrange it so that it is easier to read its meaning
- Collecting Terms
 - Very often you just need to collect like terms together
 - Unlike terms cannot be added or subtracted, like $(4x + 4y)$ remain as it is
 - Like terms can be added or subtracted, like $(4x + 4x = 8x)$
 - Like and unlike terms: $x, y, z, x^2, xy, yz, x^2y, 2x, 3y, 4xy, 12x^2$
 - Here like terms are – (x) And $(2x)$ or (y) And $(3y)$ or (xy) And $(4xy)$ or (x^2) And $(12x^2)$
 - Here unlike terms are – (x) And (y) , (x) And (z) , (y) And (z) , (x) And (x^2) , (z) And (xy) , (xy) And (yz) etc
- Removing Brackets
 - Sometimes you need to remove brackets before collecting like terms together
 - Terms outside the bracket called factor, when remove bracket then multiplies each term of inside the bracket with the factor
- Factorization
 - Rewrite an expression as the **product** of two or more numbers or expressions, and is called **factor**.
 - This usually involves using **brackets** and is called **factorization**
 - Factorization may make an expression easier to use and help to interpret its meaning
- Multiplications
 - Multiplication like $(axa = a^2)$ or $(a \times b = ab)$
- Fractions
 - The rules for working fractions in algebra are exactly the same as those used in arithmetic
 - As in arithmetic, start by finding the common denominator
 - Then write each part as the equivalent fraction with that common denominator
 - If parts are related by multiplication of the expression, then term may be cancelled top and bottom as in arithmetic
 - When the numerator (top) and/or the denominator (bottom) are not factorized, first factorize them as much as possible to get common factors which can be cancelled

2. Linear Equations

- Finding the number of sides in a polygon with an angle sum of (1) 180 degree and (2) 1080 degree

- Example 1.13:

- i. Equation: $S = 180(n - 1)$

- ii. Identity: $5(x - 1) = 5x - 5$

To distinguish an identity from an equation, the symbol ($\equiv$) is sometimes used

- iii. Steps to solve an equations

- The equation is

$$S = 180(n - 2)$$

- Submitting the value (180) for (S) gives

$$180 = 180(n - 2)$$

- Divided both sides by (180)

$$1 = 1(n - 2)$$

- Adding (2) to both sides

$$1 + 2 = (n - 2) + 2$$

- Simplify

$$3 = n - 2 + 2$$

- Tidy Up

$$3 = n$$

- Example 1.14: $5(x - 3) = 2(x + 6)$

- Open the brackets

$$5x - 15 = 2x + 12$$

- Subtract (2x) from both side

$$5x - 2x - 15 = 2x - 2x + 12$$

- Tidy Up

$$3x - 15 = 12$$

- Add (15) to both side

$$3x - 15 + 15 = 12 + 15$$

- Tidy Up

$$3x = 27$$

- Divided both side by (3)

$$\frac{3x}{3} = \frac{27}{3}$$

$$x = 9$$

- Tidy Up

- Example 1.15: $\frac{1}{2}(x + 6) = x + \frac{1}{3}(2x - 5)$

- Start by clearing the fraction.

- Multiply both sides by 6 #

$$6x \frac{1}{2}(x + 6) = 6x + 6x \frac{1}{3}(2x - 5)$$

- Tidy up

$$3(x + 6) = 6x + 2(2x - 5)$$

- Open the brackets

$$3x + 18 = 6x + 4x - 10$$

- Subtract both side by (6x), (4x) and 18

$$3x - 6x - 4x + 18 - 18 = 6x + 4x - 10 - 6x - 4x - 18$$

- Tidy up

$$-7x = -28$$

- Divided both side by (-7) #

$$\frac{-7x}{-7} = \frac{-28}{-7}$$

$$x = 4$$

- Tidy up

- Exercises

3. Changing the Subject of a Formula

- The area of a trapezium is given by $A = \frac{1}{2}(a + b)h$
- Here (a) and (b) the length of the parallel sides and (h) is the distance between them
An **equation** like this is often called a **formula**
The variable (A) is called the subject of this formula because it only appears once on its own on the left-hand side
- Often need to make one of the other variables the subject of a formula.
In this case, the steps involved are just the same as those in solving an equation
- Example 1.16: Make (a) the subject in $A = \frac{1}{2}(a + b)h$
 - It is usually easiest if you start by arranging the equation so that the variable you want to be its subject is on the left-hand side
 - Start by clearing the fraction.
 - Multiply both sides by (2) # $2A = 2 \times \frac{1}{2}(a + b)h$
 - Tidy up $2A = (a + b)h$
 - Divided both side by (h) $\frac{2A}{h} = \frac{(a+b)h}{h}$
 - Tidy up $\frac{2A}{h} = a + b$
 - Subtract both side by (b) $\frac{2A}{h} - b = a + b - b$
 - Tidy up $\frac{2A}{h} - b = a$
 - Rearrange $a = \frac{2A}{h} - b$
- Example 1.17: Make (T) the subject in the simple interest formula $I = \frac{PRT}{100}$
 - Multiply both side by (100) $I100 = \frac{PRT}{100} \times 100$
 - Tidy up $I100 = PTR$
 - Divided both side by (P) and (R) $I \frac{100}{PR} = \frac{PRT}{PR}$
 - Tidy up $I \frac{100}{PR} = T$
 - Rearrange $T = I \frac{100}{PR}$
- Example 1.18: Make (x) the subject in the formula $v = w\sqrt{a^2 - x^2}$. Speed of an Oscillating point
 - Square both side $v^2 = w^2(a^2 - x^2)$
 - Divided both side by (w²) $\frac{v^2}{w^2} = \frac{w^2(a^2 - x^2)}{w^2}$
 - Tidy up $\frac{v^2}{w^2} = a^2 - x^2$
 - Add both side by (x²) $\frac{v^2}{w^2} + x^2 = a^2 - x^2 + x^2$
 - Tidy up $\frac{v^2}{w^2} + x^2 = a^2$
 - Subtract both side by (v²/w²) $\frac{v^2}{w^2} - \frac{v^2}{w^2} + x^2 = a^2 - \frac{v^2}{w^2}$
 - Tidy up $x^2 = a^2 - \frac{v^2}{w^2}$
 - Take the root of both side $x = \sqrt{a^2 - \frac{v^2}{w^2}}$
- Example 1.19: Make (m) the subject of the formula $mv = I + mu$. Momentum after an Impulse
 - Collecting terms in (m) on left-hand side $mv - mu = I$
 - Factorise the left-hand side $m(v - u) = I$
 - Divided both side by (v - u) $\frac{m(v-u)}{(v-u)} = \frac{I}{(v-u)}$
 - Tidy up $m = \frac{I}{(v-u)}$

4. Quadratic Equations

- Example (quadratic equation)
 - The length of a rectangular field is $40m$ greater than its width
 - And its area is $6000m^2$
 - Form an equation involving the length, $(x)m$, of the field
 - Solution –
 - Length of the area is $(x)m$
 - Width of the area is $(x - 40)m$
 - The area is $m^2 = x(x - 40)$
 - So, $6000 = x(x - 40)$
 - So, $6000 = x^2 - 40x$
 - So, $6000 - 6000 = x^2 - 40x - 6000$
 - So, $0 = x^2 - 40x - 6000$
 - Equation: $x^2 - 40x - 6000 = 0$

 - This equation, involving terms in (x^2) and (x) as well as a (constant) term, is an example of **Quadratic Equation**
 - A **Linear Equation** in the variable (x) involves only terms (x) and (constant) terms
 - It is usual to write a quadratic equation with the right-hand side equal to (zero)
 - To solve, first factorize the left-hand side if possible

 - Solution $x^2 - 40x - 6000 = 0$
 - $x^2 + 60x - 100x - 6000 = 0$
 - $x(x + 60) - 100(x + 60) = 0$
 - $(x + 60)(x - 100) = 0$
 - So, $x + 60 = 0$
 - Or, $x - 100 = 0$
 - So, $x = -60$ (length cannot be negative)
 - Or, $x = 100$
 - So, length $(x) = 100m$
 - And width $(x - 40) = (100 - 40)m = 60m$
- Example 1.20: $xa + xb + ya + yb$ (a linear expression and factorization)
 - $x(a + b) + y(a + b)$
 - $(a + b)(x + y)$

 - The expression is now in the form of two factors, $(x + y)$ and $(a + b)$, this is answer
 - The same pattern is used for Quadratic Factorization.
 - But first you need to split the middle term into two parts and gives you four terms
- Example 1.21: $x^2 + 7x + 12$ (quadratic expression with all positive terms and factorization)
 - $x^2 + 4x + 3x + 12$
 - $x(x + 4) + 3(x + 4)$
 - $(x + 4)(x + 3)$

 - How to split the middle term, $7x$, into $4x + 3x$, rather than say $5x + 2x$ or $9x - 2x$
 - Here, Middle term coefficient (7) and the last term constant (12), so should split the middle terms such the way, that produce the same value of middle coefficient by adding them and same as the constant by multiplication

- Example 1.22: $x^2 - 2x - 24$ (quadratic expression with all negative terms and factorization)
 - $x^2 - 6x + 4x - 24$
 - $x(x - 6) + 4(x - 6)$
 - $(x - 6)(x + 4)$
 - It makes no difference in case of factorization of $(+4x - 6x)$ or $(-6x + 4x)$
 - $x^2 - 2x - 24$
 - $x^2 + 4x - 6x - 24$
 - $x(x + 4) - 6(x + 4)$
 - $(x + 4)(x - 6)$
 - Above both are clearly same
 - To get back to the original expression for checking, multiply the final answer
 - $(x + 4)(x - 6)$
 - $x^2 - 6x + 4x - 24$
 - $x^2 - 2x - 24$
- Example 1.23: $x^2 - 20x + 100$
 - $x^2 - 10x - 10x + 100$
 - $x(x - 10) - 10(x - 10)$
 - $(x - 10)(x - 10)$
 - $(x - 10)^2$
 - The above expression was a perfect square and help to recognize the form of such expression $(x - y)^2$, here, $y = 10$
 - Equation: $(x - y)^2 = x^2 - 2xy + y^2$
 - $x^2 - 2(x)(10) + 10(2)$
 - So $x^2 - 20x + 100$
- Example 1.25: $x^2 - 49$
 - $x^2 - 49$ can be written as $x^2 + 0x - 49$
 - $x^2 + 7x - 7x - 49$
 - $x(x + 7) - 7(x + 7)$
 - $(x + 7)(x - 7)$
 - The above expression is an example of the difference of two squares which may be written in more general form a $(x^2 - y^2)$. If here $y = 7$, then the form will be $(x^2 - y^2)$
 - Equation: $x^2 - y^2 = (x + y)(x - y)$
 - $x^2 - y^2$
 - $x^2 - 7(2)$
 - $x^2 - 49$
- Example 1.26: $6x^2 + x - 12$
 - $6x^2 - x - 12$ (first multiply the first coefficient and the constant, $12 \times 6 = 72$)
 - $6x^2 - 9x + 8x - 12$ (now factorize)
 - $3x(2x - 3) + 4(2x - 3)$
 - $(2x - 3)(3x + 4)$
 - All the above examples are factorize using this method. There coefficient of the variable (x^2) was (1) so multiplication with (1) and (constant) has not effect

- Example :
 - Before starting the procedure for factorizing a quadratic, you should always check that the terms do not have a common factor as for example in $2x^2 - 8x + 6$
 - This can be written as $2(x^2 - 4x + 3)$
 - Factorize $2(x^2 - x - 3x + 3)$
 - $2\{x(x - 1) - 3(x - 1)\}$
 - $2\{(x - 1)(x - 3)\}$
 - $2(x - 1)(x - 3)$

 - Or $2x^2 - 8x + 6$ (check)
 - $2x^2 - 6x - 2x + 6$
 - $2x(x - 3) - x(x - 3)$
 - $(x - 3)(2x - x)$

5. Solving Quadratic Equations

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- Example 1.27 – 1.29

6. Equations that Cannot be Factorized

- Introduction
- Graphical Solution
- Completing the Square
- Example 1.30

7. The Graphs of Quadratic Function

- 7i9/qaxc v/
- Introduction and Explanation
- Example 1.31

8. The Quadratic Formula

- Introduction and Explanation
- Example 1.32 – 1.33

9. Simultaneous Equations

- Introduction
- Example 1.34 – 1.36

10. Inequalities

- Linear Inequalities
- Example 1.37 – 1.40

Algebra – P2

1. Operations with Polynomials
2. Solutions of Polynomial Equations
3. The Modulus Function

Further Algebra – P3

4. The General Binomial Expansion
5. Review of Algebraic Fractions
6. Partial Fractions
7. Using Partial Fractions with the Binomial Expansion

G2: Series and Functions

Sequences and Series

1. Definitions and Notation
2. Arithmetic Progressions
3. Geometric Progressions
4. Binomial Expression

Functions

1. The Language of Functions
2. Composite Functions
3. Inverse Functions

G3: Co-Ordinate Geometry

Co-ordinate Geometry

1. Co-ordinates
2. Plotting, Sketching and Drawing
3. The Gradient of a Line
4. The Distance Between Two Points
5. The Mid-point of a Line Joining Two Points
6. The Equation of a Straight Line
7. The Intersection of Two Lines
8. Drawing Curves
9. The Intersection of a Line and a Curve

G4: Trigonometry

Trigonometry – P1

1. Trigonometry Background
2. Trigonometrical Functions
3. Trigonometrical Functions for Angles of Any Size
4. The Sine and Cosine Graphs
5. The Tangent Graph
6. Solving Equations Using Graphs of Trigonometrical Functions
7. Circular Measure
8. The Length of an Arc of a Circle
9. The Area of a Sector of a Circle
10. Other Trigonometrical Functions

Trigonometry – P2

11. Reciprocal Trigonometrical Functions
12. Compound-angle Formulae
13. Double-Angle Formulae
14. The Forms $r \cos(\theta \pm \alpha)$, $r \sin(\theta \pm \alpha)$
15. The General Solutions of Trigonometrical Equations

G5: Calculus

Differentiation – P1

1. The Gradient of a Curve
2. Finding the Gradient of a Curve
3. Finding the Gradient from first Principles
4. Differentiating by Using Standard Results
5. Using Differentiation
6. Tangents and Normals
7. Maximum and Minimum Points
8. Increasing and Decreasing Functions
9. Points of Inflection
10. The Second Derivative
11. Applications

12. The Chain Rule

Differentiation – P2

13. The Product Rule
14. The Quotient Rule
15. Differentiating Natural Logarithms and Exponentials
16. Differentiating Trigonometrical Functions
17. Differentiating Functions Defined Implicitly
18. Parametric Equations
19. Parametric Differentiation

Integration – P1

1. Reversing Differentiation
2. Finding the Area Under a Curve
3. Area as the Limit of a Sum
4. Areas Below the X Axis
5. The Area Between Two Curves
6. The Area Between a Curve and the Axis
7. The Reverse Chain Rule
8. Improper Integrals
9. Finding Volumes by Integration

Integration – P2

10. Integrals Involving the Exponential Function
11. Integrals Involving the Natural Logarithm Function
12. Integrals Involving Trigonometrical Functions
13. Numerical Integration

Further Integration – P3

14. Integration by Substitution
15. Integrals Involving Exponentials and Natural Logarithms
16. Integrals Involving Trigonometrical Functions
17. The Use of Partial Fractions in Integration
18. Integration by Parts
19. General Integration

G6: Others

Logarithms and Exponentials – P2

1. Logarithms
2. Exponential Functions
3. Modelling Curves
4. The Natural Logarithm Function
5. The Exponential Function

Numerical Solution of Equations – P2

1. Interval Estimation – Change-of-Sign Methods
2. Fixed-Point Iteration

Complex Numbers – P3

1. The Growth of the Number System
2. Working with Complex Numbers
3. Representing Complex Numbers Geometrically
4. Sets of Points in an Argand Diagram
5. The Modulus-Argument form of Complex Numbers
6. Sets of Points Using the Polar Form
7. Working with Complex Numbers in Polar Form
8. Complex Exponents
9. Complex Numbers and Equations

Differential Equations – P3

1. Forming Differential Equations from Rates of Change
2. Solving Differential Equations

G7: Vectors

Vectors – P1

1. Vectors in Two Dimensions
2. Vectors in Three Dimensions
3. Vector Calculations
4. The Angle Between Two Vectors

Vectors – P2

5. The Vector Equation of a Line
6. The Vector Equation of Two Lines
7. The Angle Between Two Lines
8. The Perpendicular Distance from a Point to a Line
9. The Vector Equation of a Plane

10. The Intersection of a Line and a Plane
11. The Distance of a Point from a Plane
12. The Angle Between a Line and a Plane
13. The Intersection of Two Planes